

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 32

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Most sulfur in coal is present as (solid) iron pyrites (FeS_2) (1) • Coal is crushed and put in a stream of flowing water: the coal is carried away while the denser pyrites are left behind (washing / streaming) (1)	2
01.2	<i>1 mark for each point.</i> • Flue gases are passed through a bed of crushed calcium carbonate (limestone) (1) • Sulfur dioxide reacts (with oxygen and calcium carbonate) to form solid calcium sulfate (and carbon dioxide) (1) • If the gases have already been cleaned of smoke, the calcium sulfate is pure enough to be sold as gypsum for building plaster (1)	3
01.3	<i>1 mark for each point.</i> • The sodium hydrogen sulfite formed can be heated to regenerate sodium sulfite (re-used) and release concentrated sulfur dioxide (1) • The pure sulfur dioxide is a valuable raw material for making sulfur / sulfuric acid, so a waste becomes a resource (1)	2
01.4	<i>1 mark for each point.</i> • Fluidised bed: increased surface area allows rapid combustion at lower temperatures so less nitrogen and oxygen react to form NO_x (production prevented) (1) • Catalytic converter: platinum / palladium catalyst chemically reduces NO_x back to nitrogen and oxygen gases in the exhaust (1) • Urea sprayed into the hot flue / exhaust gases reacts with NO_x, converting it to nitrogen, carbon dioxide and water (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 252-253; AQA spec 3.4.3.2.2-3.4.3.2.3.	

Question	Marking guidance	Marks
02.1	<i>1 mark.</i> • The time taken for half of a pollutant (released into the environment) to break down / degrade (1)	1
02.2	<i>1 mark for each point.</i> • DDT is liposoluble so it is stored in fat / can pass through phospholipid cell membranes and is not excreted (nitrates are water soluble and excreted easily) (1) • DDT is persistent (long half-life) so it is not broken down before it is passed to the next trophic level (1) • So DDT bioaccumulates in each organism and becomes more concentrated at each successive trophic level (1)	3
02.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • The gas is denser than air so it settled close to the ground / did not rise and disperse (1) • Low wind velocity (at night) meant it was not dispersed / diluted (1) • Short half-life so it broke down before travelling far (1)	2
02.4	<i>1 mark for each point.</i> • High solubility means nitrates are easily dispersed / leached / carried by water so they affect a larger area (eg rivers, aquifers, the sea) (1) • But dispersal dilutes the pollutant so the concentration at any one point may be lower (1)	2
02.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Noise / sound (1) • Heat / thermal pollution (1) • Ionising radiation (1) • Light (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	
03.1	<i>1 mark for each point.</i> • Gravimetry: gravimeters detect variations in gravity caused by differences in rock density / mass (metal ores usually denser than sedimentary rock) (1) • Magnetometry (1) • Seismic survey: sound / shock waves from controlled explosions or a seismic vibrator; echoes recorded by geophones give the depth, density and shape of rock strata (1) • Resistivity (1)	4
03.2	<i>1 mark for each point.</i> • Trial drilling is the most expensive technique per site (1) • Remote sensing (satellite, aircraft, ground surveys) covers large areas at low cost per unit area and identifies promising sites (1) • Drilling is then targeted where deposits are likely; it is the only method that produces actual rock samples for chemical analysis to confirm composition and purity (proven reserves) (1)	3
03.3	<i>1 mark for each point.</i> • Different minerals emit / reflect infrared at different wavelengths (1) • Multispectral sensors detect these wavelengths over the whole surface (eg Landsat 8 every 16 days) giving information on surface mineral composition (1)	2

Question	Marking guidance	Marks
03.4	1 mark. Results are available immediately in the field, saving time waiting for laboratory analysis (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 137; AQA spec 3.2.3.4.	
04.1	1 mark for each point. - Instability / landslips on the hillside, especially after heavy rain - compaction, drainage, establishing vegetation whose roots bind the spoil, regrading slopes (1) - Acid drainage from oxidation of pyrites, with toxic metals - collect and treat leachate, add lime to raise pH (1) - Spontaneous combustion of residual coal / hydrocarbons - layers of fine-grained material to reduce air flow (1) - Lack of nutrients / poor plant growth - add topsoil, sewage sludge, fertiliser (1) - Unattractive topography - landscaping / reshaping (1) - Dust / heavy metals - cover and keep at high pH (1)	6
04.2	1 mark for each point. - So that plant roots cannot reach the toxic layer (1) - Otherwise toxins could be taken up and brought to the surface / dispersed via leaf litter and food chains (1)	2
04.3	1 mark for each point. - Steel slag is alkaline, producing unusual calcareous / high-pH soil conditions (1) - Supports uncommon plants such as cowslips / orchids that cannot compete on fertile neutral soils (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 282-283; AQA spec 3.4.3.2.12.	
05.1	1 mark for each point. - Quadrat size depends on the size / spacing of the organisms: 0.5 m x 0.5 m is typical for ground flora (1) 10 cm x 10 cm is for lichens (too small to contain a representative sample of plants); 100 m x 100 m is for trees (impractical, too much work) (1)	2
05.2	A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown. $N = 240$; $N(N-1) = 240 \times 239 = 57\,360$ (1) $\Sigma n(n-1) = 48 \times 47 + 120 \times 119 + 36 \times 35 + 24 \times 23 + 12 \times 11 = 2256 + 14280 + 1260 + 552 + 132 = 18\,480$ (1) $D = 57\,360 / 18\,480 = 3.10$ (accept 3.1) (1)	3
05.3	1 mark for each point. - Lower D means lower biodiversity: the ungrazed meadow is dominated by one or a few species (eg tall grass) with the others rare (less even) (1) - Grazing (plagioclimax management) prevents dominance by vigorous species so abundance is more evenly spread (1)	2
05.4	1 mark for each point. - A frame quadrat flattens tall plants, making percentage cover estimates inaccurate and hiding shorter plants (1) - Point quadrat pins are lowered through the vegetation; each touch counts (10 pins = 10% each) without flattening (1)	2

Question	Marking guidance	Marks
05.5	<i>1 mark.</i> Subjective judgement varies between recorders; results are not quantitative so statistical analysis is not possible (but quick for many areas) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-395, 401-402; AQA spec 3.7.2.1.	
06.1	<i>1 mark for each point.</i> - No large dam / reservoir needed so no flooding of farmland, homes or habitats and no barrier across the whole river (run-of-river / micro-hydro diverts part of the flow) (1) - Suitable for rivers with a small drop in height and lower flow rates that are common in lowland rural areas (1) - Small scale matches local demand; electricity used locally without long transmission lines / grid connection (1) - Low construction cost and simple technology maintained by the community (1) - Helical turbines tolerate high turbidity / debris without damage and fish pass through unharmed; Kaplan turbines with adjustable blades cope with variable flow (1)	5
06.2	<i>1 mark.</i> Lack of capital / finance, technical knowledge or grid access; seasonal drying of rivers; conflicts with irrigation water; planning / land ownership (1)	1
06.3	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> - Output is proportional to flow x efficiency: $100 \times 0.10 \times (40 / 90)$ (1) = 4.4 kW (accept 4.4-4.5) (1)	2
06.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Constant regulated flow removes seasonal extremes: dry-season sandbanks used by nesting birds / turtles lost; high flows that clear gravel spawning beds lost; floodplain no longer flooded (1) - Sudden flow increases for peak generation raise turbidity, wash away species, reduce light for photosynthesis (1) - Sediment trapped so less nutrient replenishment downstream; barrier to migrating fish (salmon, sturgeon) and isolation of populations (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 203-205; AQA spec 3.3.4.1.	
07.1	<i>Correct answer only.</i> Organic (methyl) mercury (1)	1
07.2	<i>1 mark for each point.</i> - Anaerobic microbes converted the inorganic mercury into organic methyl mercury (1) - Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people (1)	2
07.3	<i>1 mark for each point.</i> - Most heavy metals are less soluble at high pH (1) - So they remain immobile and are not leached into water or absorbed by organisms (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	

Question	Marking guidance	Marks
08.1	<i>1 mark for each point.</i> • B (about 40:40:20 sand:silt:clay) (1) • Balanced mix gives good drainage and aeration from the sand fraction (1) • Water retention and nutrient (ion) retention from the clay fraction; easy root penetration and cultivation (1)	3
08.2	<i>1 mark for each point.</i> • Large sand particles leave large pore spaces so water drains rapidly (high permeability / aeration) (1) • Nutrient ions adsorb onto the (charged) surfaces of clay particles but not onto sand (1) • With only 5% clay there is little surface to hold ions so nutrients are leached; low water retention too (1)	3
08.3	<i>1 mark for each point.</i> • C (clay) holds much water so has a high thermal capacity: warms and cools slowly; A warms rapidly after cold weather (1) • Clay particles held together by capillary water so roots find it hard to push between them; penetration is easy in sandy A (1)	2
08.4	<i>1 mark for each point.</i> • Shake a known mass of soil with water in a measuring cylinder and leave to settle; particles settle in order of size (sand first, then silt, then clay; organic matter floats) (1) • Measure the depth / volume of each layer to calculate the percentage of each fraction, then use the soil triangle to name the textural class (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-160, 412; AQA spec 3.2.5.1.	
09.1	<i>Up to 6 marks in total; award marks in proportion to the completeness of each point.</i> • Habitat change: flooding destroys the previous terrestrial / river habitats but creates a wetland which may be valuable (wetlands are uncommon) (up to 2) • Wildlife barrier: the dam blocks migrating fish (salmon, sturgeon) and prevents recolonisation along the river (up to 2) • Changed river regime downstream: regulated flow removes low-flow periods needed by species such as river turtles nesting on sandbanks and high flows that wash sediment from gravel spawning beds (up to 2) • Sedimentation: sediment settles in the reservoir instead of fertilising the floodplain and building river banks / coastlines downstream, increasing erosion (up to 2) • Microclimate: water's high heat capacity reduces temperature fluctuations; higher wind speeds; more evaporation, humidity, cloud and precipitation downwind (up to 2)	6
09.2	<i>1 mark for each point.</i> • Vegetation / dead organic matter under water decays anaerobically (1) • Releasing methane, a powerful greenhouse gas (1)	2
09.3	<i>1 mark for each point.</i> • After heavy rain surplus water is stored, reducing peak flow / flooding downstream (1) • In dry periods the dam releases water to maintain river levels / supplies (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117; AQA spec 3.2.2.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Organic nutrients (sewage, manure, silage effluent, paper mill / tannery / food processing effluent) are food for aerobic microbes whose respiration deoxygenates the water, killing fish and invertebrates; bacterial growths cover the bed, blocking light • Pathogens in sewage (cholera, typhoid, dysentery) spread via contaminated water; decay of organic matter releases inorganic nutrients • Inorganic nutrients: nitrates (fertiliser / manure runoff, leaching) and phosphates (sewage effluent, detergents, eroded soil) cause cultural eutrophication: algal blooms, shading of macrophytes, cyanobacterial toxins, night-time and end-of-summer deoxygenation • Human health: nitrites and methaemoglobinaemia; possible carcinogen via nitrosamines • Control of organic wastes: sewage treatment (pre-treatment, primary sedimentation removing >95% organic matter, aeration / activated sludge or trickling filters, anaerobic digestion of sludge with biogas) • Control of nutrients: tertiary phosphate stripping with iron(III) sulfate, denitrification in filter beds, micro-strainers / UV; Nitrate Vulnerable Zones, fertiliser timing / amount, low-solubility fertilisers, legumes, manure management; dredging sediments • Evaluation: sewage treatment has virtually eliminated deoxygenation and waterborne disease in developed countries but storm overflows and combined sewers still discharge untreated sewage; tertiary treatment is expensive so only used at sensitive sites; diffuse agricultural nitrate is hard to control and aquifers respond slowly; phosphate legacy in sediments • Judgement on point-source (sewage) control being more successful than diffuse-source (agriculture) control	9
	Model answer - Level 3 (9 marks)	
	Organic wastes - sewage, manure, silage effluent, paper-mill and food-processing effluent - are food for aerobic bacteria, whose respiration strips dissolved oxygen from the water so fish and invertebrates suffocate and bacterial growths smother the bed; sewage also carries pathogens such as cholera and typhoid. Inorganic nutrients act differently: nitrate leached from fertiliser and manure and phosphate from sewage, detergents and eroded soil cause cultural eutrophication, in which algal blooms shade macrophytes, cyanobacteria release toxins and night-time respiration and decay deoxygenate the water. Nitrate in drinking water can cause methaemoglobinaemia and may form carcinogenic nitrosamines.	

Question	Marking guidance	Marks
	Point sources are controlled by sewage treatment: screening and grit removal, primary sedimentation removing most organic solids, aeration with activated sludge or trickling filters, and anaerobic digestion of sludge yielding biogas. Nutrients need tertiary treatment - phosphate stripping with iron(III) sulfate, denitrifying filter beds, micro-strainers and UV. Diffuse agricultural sources are managed by Nitrate Vulnerable Zones, limits on the timing and amount of fertiliser, low-solubility fertilisers, legumes, manure management and buffer strips.	
	Sewage treatment has virtually eliminated river deoxygenation and waterborne disease in developed countries, but storm overflows on combined sewers still discharge untreated sewage, and tertiary treatment is expensive so it is used only where eutrophication is a risk. Diffuse nitrate from farmland is much harder to control because it comes from thousands of sources and depends on farmers' behaviour and the weather. Control of point sources has therefore been far more successful than control of diffuse ones.	
	Weaker answer - Level 1 (3 marks)	
	Sewage in rivers uses up the oxygen because bacteria break it down, so fish die. Fertilisers wash into rivers and cause eutrophication, where algae grow and block the light and then the water runs out of oxygen. Sewage is treated in sewage works to clean it before it goes into rivers. Farmers should use less fertiliser. Sewage treatment works well but fertiliser is still a problem.	
	What makes the difference	
	• Level 1 has the outline of both mechanisms but Level 3 explains them precisely (aerobic respiration, shading, night-time respiration, pathogens, health effects) and names the pollutant sources.	
	• Level 3 describes the stages of sewage treatment and the specific nutrient controls; Level 1 says 'sewage works' and 'less fertiliser'.	
	• Level 3's evaluation gives reasons (storm overflows, cost of tertiary treatment, diffuse sources) for its point-vs-diffuse judgement; Level 1 asserts it.	
10.2	<i>1 mark.</i> • Algae / cyanobacteria absorb nutrients directly from the water (not the sediments) so they use dissolved nutrients more easily and grow rapidly (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-272; AQA spec 3.4.3.2.9.	

Question	Marking guidance	Marks
11.1	Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Features of fertile soil: water, nutrients, air, organic matter, pH, biota, texture, structure, depth; slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • Causes of degradation: vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • Types of erosion: rain splash, runoff, wind, slumping; consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • Management: long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy • Evaluation and judgement on which strategies are most effective in different environments Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	25

Question	Marking guidance	Marks
11.2	Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Negative feedback: increased low-level cloud (albedo), increased photosynthesis and sequestration; hydrological cycle evaporation-precipitation • Positive feedback: melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • Tipping points: self-sustaining processes; why control must precede them; examples • Implications: urgency and precautionary principle; need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation; geoengineering risks • Difficulty of prediction because feedbacks interact at different scales; models; international agreements and per capita footprints • Judgement on whether current strategies are adequate Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	25